

LEVEL 8: Issue Book

1. Objective

Issue a selected book to a student only when the book exists and is currently available.

2. Data Structure / Algorithm Used

Array + Linear Search + State Update

3. Problem Statement

Extend the Library Management System by implementing the issue book module. Complete this module independently before integrating it with previous levels.

4. Algorithm / Working

Read student and book IDs → search the book → check availability → change availability to issued.

5. C Program

```
#include <stdio.h>
typedef struct { int id; char title[40]; int available; } Book;

int search(Book b[],int n,int id) {
    int i; for(i=0;i<n;i++) if(b[i].id==id) return i;
    return -1;
}

int main() {
    Book b[]={101,"C Programming",1},{105,"Data Structures",1},
             {110,"Operating System",0}};
    int studentID,bookID,pos;
    printf("Student ID: "); scanf("%d",&studentID);
    printf("Book ID: "); scanf("%d",&bookID);
    pos=search(b,3,bookID);

    if(pos==-1) printf("Book not found\n");
    else if(!b[pos].available) printf("Book is already issued\n");
    else {
        b[pos].available=0;
        printf("Book %d issued to student %d\n",bookID,studentID);
    }
    return 0;
}
```

6. Sample Input

Student ID: 201
Book ID: 105

7. Sample Output

Book 105 issued to student 201

8. Complexity

Book search: $O(n)$. Availability update after finding: $O(1)$.

9. Practice Exercises

- Store issued student ID.
- Implement Return Book.
- Prevent issuing an unavailable book.

10. Related Viva Questions

- What checks are needed before issuing?
- Why store availability?
- What happens if the book does not exist?
- What is the search complexity?